Retail Sales Analysis - SQL Project

Overview

This project focuses on analyzing retail sales data using SQL queries. It aims to explore customer behavior, sales trends, and key business insights through structured database queries. The dataset consists of transaction details, including product categories, customer demographics, and sales amounts.

Questions and Purpose

1. How many unique customers made a purchase?

Purpose: Helps businesses understand customer reach and potential retention strategies.

2. What are the top-selling product categories?

Purpose: Identifies the best-performing products, guiding inventory and marketing strategies.

3. Which month had the highest sales?

Purpose: Determines seasonal trends, helping businesses optimize stock and promotions.

4. What is the average purchase amount per transaction?

Purpose: Provides insights into customer spending behavior and pricing strategy effectiveness.

5. How do sales vary across different times of the day?

Purpose: Helps in staffing and operational decisions to maximize efficiency during peak hours.

6. Who are the top 5 customers based on total purchases?

Purpose: Identifies high-value customers for loyalty programs and personalized marketing.

7. How do male and female customers differ in their purchasing habits?

Purpose: Helps businesses tailor marketing efforts based on gender-based buying trends.

8. What is the total sales revenue per category?

Purpose: Evaluates the contribution of each category to overall revenue.

9. Are there high-value transactions (above a certain threshold)?

Purpose: Highlights premium purchases and potential upsell opportunities.

10. How many transactions had missing or null values?

Purpose: Ensures data accuracy and completeness before performing deeper analysis.

Conclusion

This project provides valuable insights into retail sales data by leveraging SQL queries for trend analysis, customer segmentation, and business decision-making. The findings can assist businesses in optimizing inventory, marketing strategies, and customer engagement.